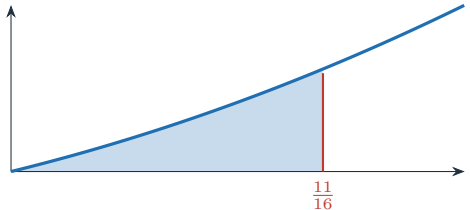
Условие

Дадена е X с плътност $f_X(x) = c(x^2 + 2x)\mathbb{1}_{[0,1]}(x)$. Намерете (а) c; (б) $\mathbb{E}[X]$ и $\mathrm{Var}(X)$; (в) $\mathbb{P}(X \leq \mathbb{E}[X])$; (г) $\mathbb{E}[X^2 + 3X]$.

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(а) $\int_0^1 c(x^2+2x)dx = \frac{4}{3}c = 1$, значи $c = \frac{3}{4}$.

(б) $\mathbb{E}[X] = \frac{3}{4}\int_0^1 (x^3+2x^2)dx = \frac{11}{16}$ и $\mathbb{E}[X^2] = \frac{21}{40}$, откъдето $\mathrm{Var}(X) = \frac{67}{1280}$.

(в) Защрихованото лице: $\frac{3}{4}\left(\frac{x^3}{3}+x^2\right)\Big|_{x=11/16} \approx 0.436$.

(г) $\frac{21}{40} + 3 \cdot \frac{11}{16} = \frac{207}{80}$.

Отговор. $c = \frac{3}{4}$; $\frac{11}{16}$, $\frac{67}{1280}$; ≈ 0.436; $\frac{207}{80}$.